

**10**      **B**      Energy required per second to raise water

$$= mgh / t = (V)gh / t$$

$$= (1000) (0.50) (9.81) (30) / (60)$$

$$= 2.45 \text{ kW}$$

As the pump is only 70% efficient, the engine's power =  $2.45 / 0.70 = 3.5 \text{ kW}$

**11**      **B**      The net force is the centripetal force =  $mr\omega^2$